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VV156 Final RC Part I

Integration

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Contents

- Integral Techniques
- Improper Integral

1 Table of Indefinite Integrals

$$\int cf(x) dx = c \int f(x) dx$$

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

$$\int k dx = kx + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \frac{1}{x^2 + 1} dx = \tan^{-1} x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

$$\int \sinh x dx = \cosh x + C$$

$$\int \cosh x dx = \sinh x + C$$



Properties

1 Linearity

$$\int_a^b [k_1 f(x) + k_2 g(x)] dx = k_1 \int_a^b f(x) dx + k_2 \int_a^b g(x) dx$$

2 Internal Additivity

$$\int_a^c f(x) dx + \int_c^b f(x) dx = \int_a^b f(x) dx$$

3 Comparison Properties

If $f(x) \geq g(x)$ for $a \leq x \leq b$, then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$

If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$



Properties

④ Absolute Integrability

If $f(x)$ is integrable on $[a,b]$, then $|f(x)|$ is also integrable on $[a,b]$, and we have

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

Note

If $f(x)$ and $g(x)$ are integrable on $[a,b]$, then $f(x) \cdot g(x)$ is also integrable on $[a,b]$, but generally

$$\int_a^b f(x)g(x)dx \neq \left(\int_a^b f(x)dx \right) \cdot \left(\int_a^b g(x)dx \right)$$



4 The Substitution Rule If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I , then

$$\int f(g(x)) g'(x) dx = \int f(u) du$$

6 The Substitution Rule for Definite Integrals If g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$



Very Important!

Table of Trigonometric Substitutions

Expression	Substitution	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, \quad 0 \leq \theta < \frac{\pi}{2} \text{ or } \pi \leq \theta < \frac{3\pi}{2}$	$\sec^2 \theta - 1 = \tan^2 \theta$

EXAMPLE 6

Find $\int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx.$

EXAMPLE 7

Evaluate $\int \frac{x}{\sqrt{3 - 2x - x^2}} dx.$



SOLUTION First we note that $(4x^2 + 9)^{3/2} = (\sqrt{4x^2 + 9})^3$ so trigonometric substitution is appropriate. Although $\sqrt{4x^2 + 9}$ is not quite one of the expressions in the table of trigonometric substitutions, it becomes one of them if we make the preliminary substitution $u = 2x$. When we combine this with the tangent substitution, we have $x = \frac{3}{2} \tan \theta$, which gives $dx = \frac{3}{2} \sec^2 \theta d\theta$ and

$$\sqrt{4x^2 + 9} = \sqrt{9 \tan^2 \theta + 9} = 3 \sec \theta$$

When $x = 0$, $\tan \theta = 0$, so $\theta = 0$; when $x = 3\sqrt{3}/2$, $\tan \theta = \sqrt{3}$, so $\theta = \pi/3$.

$$\begin{aligned} \int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx &= \int_0^{\pi/3} \frac{\frac{27}{8} \tan^3 \theta}{27 \sec^3 \theta} \frac{3}{2} \sec^2 \theta d\theta \\ &= \frac{3}{16} \int_0^{\pi/3} \frac{\tan^3 \theta}{\sec \theta} d\theta = \frac{3}{16} \int_0^{\pi/3} \frac{\sin^3 \theta}{\cos^2 \theta} d\theta \\ &= \frac{3}{16} \int_0^{\pi/3} \frac{1 - \cos^2 \theta}{\cos^2 \theta} \sin \theta d\theta \end{aligned}$$

Now we substitute $u = \cos \theta$ so that $du = -\sin \theta d\theta$. When $\theta = 0$, $u = 1$; when $\theta = \pi/3$, $u = \frac{1}{2}$. Therefore

$$\begin{aligned} \int_0^{3\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx &= -\frac{3}{16} \int_1^{1/2} \frac{1 - u^2}{u^2} du \\ &= \frac{3}{16} \int_{1/2}^1 (1 - u^2) du = \frac{3}{16} \left[u + \frac{1}{3} u^3 \right]_{1/2}^1 \\ &= \frac{3}{16} \left[\left(\frac{1}{2} + 2 \right) - \left(1 + \frac{1}{24} \right) \right] = \frac{3}{32} \end{aligned}$$

SOLUTION We can transform the integrand into a function for which trigonometric substitution is appropriate by first completing the square under the root sign:

$$\begin{aligned} 3 - 2x - x^2 &= 3 - (x^2 + 2x) = 3 + 1 - (x^2 + 2x + 1) \\ &= 4 - (x + 1)^2 \end{aligned}$$

This suggests that we make the substitution $u = x + 1$. Then $du = dx$ and $x = u - 1$, so

$$\int \frac{x}{\sqrt{3 - 2x - x^2}} dx = \int \frac{u - 1}{\sqrt{4 - u^2}} du$$

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We now substitute $u = 2 \sin \theta$, giving $du = 2 \cos \theta d\theta$ and $\sqrt{4 - u^2} = 2 \cos \theta$, so

$$\begin{aligned} \int \frac{x}{\sqrt{3 - 2x - x^2}} dx &= \int \frac{2 \sin \theta - 1}{2 \cos \theta} 2 \cos \theta d\theta \\ &= \int (2 \sin \theta - 1) d\theta \\ &= -2 \cos \theta - \theta + C \\ &= -\sqrt{4 - u^2} - \sin^{-1}\left(\frac{u}{2}\right) + C \\ &= -\sqrt{3 - 2x - x^2} - \sin^{-1}\left(\frac{x + 1}{2}\right) + C \end{aligned}$$



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4–30 Evaluate the integral.

4. $\int_0^1 x^3 \sqrt{1-x^2} \, dx$

5. $\int_{\sqrt{2}}^2 \frac{1}{t^3 \sqrt{t^2-1}} \, dt$

7. $\int_0^a \frac{dx}{(a^2+x^2)^{3/2}}, \quad a > 0$

9. $\int \frac{dx}{\sqrt{x^2+16}}$

11. $\int \sqrt{1-4x^2} \, dx$

13. $\int \frac{\sqrt{x^2-9}}{x^3} \, dx$

15. $\int_0^a x^2 \sqrt{a^2-x^2} \, dx$

17. $\int \frac{x}{\sqrt{x^2-7}} \, dx$

19. $\int \frac{\sqrt{1+x^2}}{x} \, dx$

6. $\int_0^3 \frac{x}{\sqrt{36-x^2}} \, dx$

8. $\int \frac{dt}{t^2 \sqrt{t^2-16}}$

10. $\int \frac{t^5}{\sqrt{t^2+2}} \, dt$

12. $\int \frac{du}{u\sqrt{5-u^2}}$

14. $\int_0^1 \frac{dx}{(x^2+1)^2}$

16. $\int_{\sqrt{2}/3}^{2/3} \frac{dx}{x^5 \sqrt{9x^2-1}}$

18. $\int \frac{dx}{[(ax)^2-b^2]^{3/2}}$

20. $\int \frac{x}{\sqrt{1+x^2}} \, dx$

Just for practice

Do as you want



Very Important!

11

Integration by Parts

$$\int u dv = uv - \int v du$$
$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

How to choose $f(x)$ and $g'(x)$?

Among the following function types:

the more to the left, the more suitable to be $f(x)$

the more to the right, the more suitable to be $g'(x)$

(left) inverse trigonometric function, logarithm function, power function, trigonometric function, exponential function **(right)**

$$\int_a^b f(x)g'(x) dx = f(x)g(x)\Big|_a^b - \int_a^b g(x)f'(x) dx$$



Evaluate $\int_0^1 (1-t)^p t^q dt$

V **EXAMPLE 4** Evaluate $\int e^x \sin x dx$.

EXAMPLE 5 Calculate $\int_0^1 \tan^{-1} x dx$.

SOLUTION Neither e^x nor $\sin x$ becomes simpler when differentiated, but we try choosing $u = e^x$ and $dv = \sin x \, dx$ anyway. Then $du = e^x \, dx$ and $v = -\cos x$, so integration by parts gives

$$\boxed{4} \quad \int e^x \sin x \, dx = -e^x \cos x + \int e^x \cos x \, dx$$

The integral that we have obtained, $\int e^x \cos x \, dx$, is no simpler than the original one, but at least it's no more difficult. Having had success in the preceding example integrating by parts twice, we persevere and integrate by parts again. This time we use $u = e^x$ and $dv = \cos x \, dx$. Then $du = e^x \, dx$, $v = \sin x$, and

$$\boxed{5} \quad \int e^x \cos x \, dx = e^x \sin x - \int e^x \sin x \, dx$$

At first glance, it appears as if we have accomplished nothing because we have arrived at $\int e^x \sin x \, dx$, which is where we started. However, if we put the expression for $\int e^x \cos x \, dx$ from Equation 5 into Equation 4 we get

$$\int e^x \sin x \, dx = -e^x \cos x + e^x \sin x - \int e^x \sin x \, dx$$

This can be regarded as an equation to be solved for the unknown integral. Adding $\int e^x \sin x \, dx$ to both sides, we obtain

$$2 \int e^x \sin x \, dx = -e^x \cos x + e^x \sin x$$

Dividing by 2 and adding the constant of integration, we get

$$\int e^x \sin x \, dx = \frac{1}{2} e^x (\sin x - \cos x) + C$$



SOLUTION Let

$$u = \tan^{-1}x \quad dv = dx$$

Then

$$du = \frac{dx}{1+x^2} \quad v = x$$

So Formula 6 gives

$$\begin{aligned} \int_0^1 \tan^{-1}x \, dx &= x \tan^{-1}x \Big|_0^1 - \int_0^1 \frac{x}{1+x^2} \, dx \\ &= 1 \cdot \tan^{-1}1 - 0 \cdot \tan^{-1}0 - \int_0^1 \frac{x}{1+x^2} \, dx \\ &= \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} \, dx \end{aligned}$$

To evaluate this integral we use the substitution $t = 1 + x^2$ (since u has another meaning in this example). Then $dt = 2x \, dx$, so $x \, dx = \frac{1}{2} dt$. When $x = 0$, $t = 1$; when $x = 1$, $t = 2$; so

$$\begin{aligned} \int_0^1 \frac{x}{1+x^2} \, dx &= \frac{1}{2} \int_1^2 \frac{dt}{t} = \frac{1}{2} \ln |t| \Big|_1^2 \\ &= \frac{1}{2} (\ln 2 - \ln 1) = \frac{1}{2} \ln 2 \end{aligned}$$

Therefore

$$\int_0^1 \tan^{-1}x \, dx = \frac{\pi}{4} - \int_0^1 \frac{x}{1+x^2} \, dx = \frac{\pi}{4} - \frac{\ln 2}{2}$$



- ① If at least m and n is odd, adopt substitution rule.

$$\int \sin^m x \cos^{2n+1} x \, dx = \int \sin^m x \cos^{2n} x \, d(\sin x) =$$

$$\int \sin^m x \cdot (1 - \sin^2 x)^n \, d(\sin x) \stackrel{u=\sin x}{=} \int u^m (1 - u^2)^n \, du$$

$$\int \cos^m x \sin^{2n+1} x \, dx = - \int \cos^m x \sin^{2n} x \, d(\cos x) =$$

$$- \int \cos^m x \cdot (1 - \cos^2 x)^n \, d(\cos x) \stackrel{u=\cos x}{=} - \int u^m (1 - u^2)^n \, du$$

- ② Else, use half-angle identities that $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$,

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x).$$

- ① If m is even (not 0), use $d(\tan x) = \sec^2 x dx$ to transform it into the integration of $\tan x$.

$$\int \sec^{2k} x \tan^n x dx = \int (\tan^2 x + 1)^{k-1} \tan^n x d(\tan x) \stackrel{u=\tan x}{=} \int (u^2 + 1)^{k-1} u^n du.$$

- ② If n is odd, use $d(\sec x) = \sec x \tan x dx$ to transform it into the integration of $\sec x$.

$$\int \tan^{2k+1} x \sec^n x dx = \int (\sec^2 x - 1)^k \sec^{n-1} x d(\sec x) = \dots$$

Though not as useful, you can also explore $\int \csc^m x \cot^n x dx$.

V **EXAMPLE 2** Find $\int \sin^5 x \cos^2 x \, dx$.

EXAMPLE 6 Find $\int \tan^5 \theta \sec^7 \theta \, d\theta$.

SOLUTION We could convert $\cos^2 x$ to $1 - \sin^2 x$, but we would be left with an expression in terms of $\sin x$ with no extra $\cos x$ factor. Instead, we separate a single sine factor and rewrite the remaining $\sin^4 x$ factor in terms of $\cos x$:

$$\sin^5 x \cos^2 x = (\sin^2 x)^2 \cos^2 x \sin x = (1 - \cos^2 x)^2 \cos^2 x \sin x$$

Substituting $u = \cos x$, we have $du = -\sin x \, dx$ and so

$$\begin{aligned} \int \sin^5 x \cos^2 x \, dx &= \int (\sin^2 x)^2 \cos^2 x \sin x \, dx \\ &= \int (1 - \cos^2 x)^2 \cos^2 x \sin x \, dx \\ &= \int (1 - u^2)^2 u^2 (-du) = -\int (u^2 - 2u^4 + u^6) \, du \\ &= -\left(\frac{u^3}{3} - 2\frac{u^5}{5} + \frac{u^7}{7}\right) + C \\ &= -\frac{1}{3}\cos^3 x + \frac{2}{5}\cos^5 x - \frac{1}{7}\cos^7 x + C \end{aligned}$$



SOLUTION If we separate a $\sec^2 \theta$ factor, as in the preceding example, we are left with a $\sec^5 \theta$ factor, which isn't easily converted to tangent. However, if we separate a $\sec \theta \tan \theta$ factor, we can convert the remaining power of tangent to an expression involving only secant using the identity $\tan^2 \theta = \sec^2 \theta - 1$. We can then evaluate the integral by substituting $u = \sec \theta$, so $du = \sec \theta \tan \theta d\theta$:

$$\begin{aligned}\int \tan^5 \theta \sec^7 \theta d\theta &= \int \tan^4 \theta \sec^6 \theta \sec \theta \tan \theta d\theta \\&= \int (\sec^2 \theta - 1)^2 \sec^6 \theta \sec \theta \tan \theta d\theta \\&= \int (u^2 - 1)^2 u^6 du \\&= \int (u^{10} - 2u^8 + u^6) du \\&= \frac{u^{11}}{11} - 2 \frac{u^9}{9} + \frac{u^7}{7} + C \\&= \frac{1}{11} \sec^{11} \theta - \frac{2}{9} \sec^9 \theta + \frac{1}{7} \sec^7 \theta + C\end{aligned}$$



Use Product-to-Sum formula:

$$\textcircled{1} \sin \alpha \cos \beta = \frac{1}{2}(\sin(\alpha + \beta) + \sin(\alpha - \beta))$$

$$\textcircled{2} \cos \alpha \sin \beta = \frac{1}{2}(\sin(\alpha + \beta) - \sin(\alpha - \beta))$$

$$\textcircled{3} \sin \alpha \sin \beta = \frac{1}{2}(\cos(\alpha + \beta) + \cos(\alpha - \beta))$$

$$\textcircled{4} \cos \alpha \cos \beta = -\frac{1}{2}(\cos(\alpha + \beta) - \cos(\alpha - \beta))$$

EXAMPLE 9 Evaluate $\int \sin 4x \cos 5x \, dx$.

SOLUTION This integral could be evaluated using integration by parts, but it's easier to use the identity in Equation 2(a) as follows:

$$\begin{aligned}\int \sin 4x \cos 5x \, dx &= \int \frac{1}{2}[\sin(-x) + \sin 9x] \, dx \\ &= \frac{1}{2} \int (-\sin x + \sin 9x) \, dx \\ &= \frac{1}{2}(\cos x - \frac{1}{9} \cos 9x) + C\end{aligned}$$

$$9. \int_0^{\pi} \cos^4(2t) dt$$

$$10. \int_0^{\pi} \sin^2 t \cos^4 t dt$$

$$27. \int_0^{\pi/3} \tan^5 x \sec^4 x dx$$

$$28. \int \tan^5 x \sec^3 x dx$$

$$11. \int_0^{\pi/2} \sin^2 x \cos^2 x dx$$

$$12. \int_0^{\pi/2} (2 - \sin \theta)^2 d\theta$$

$$29. \int \tan^3 x \sec x dx$$

$$30. \int_0^{\pi/4} \tan^4 t dt$$

$$13. \int t \sin^2 t dt$$

$$14. \int \cos \theta \cos^5(\sin \theta) d\theta$$

$$31. \int \tan^5 x dx$$

$$32. \int \tan^2 x \sec x dx$$

$$15. \int \frac{\cos^5 \alpha}{\sqrt{\sin \alpha}} d\alpha$$

$$16. \int x \sin^3 x dx$$

$$33. \int x \sec x \tan x dx$$

$$34. \int \frac{\sin \phi}{\cos^3 \phi} d\phi$$

$$41. \int \sin 8x \cos 5x dx$$

$$42. \int \cos \pi x \cos 4\pi x dx$$

$$43. \int \sin 5\theta \sin \theta d\theta$$

$$44. \int \frac{\cos x + \sin x}{\sin 2x} dx$$

$$45. \int_0^{\pi/6} \sqrt{1 + \cos 2x} dx$$

$$46. \int_0^{\pi/4} \sqrt{1 - \cos 4\theta} d\theta$$

$$47. \int \frac{1 - \tan^2 x}{\sec^2 x} dx$$

$$48. \int \frac{dx}{\cos x - 1}$$

$$49. \int x \tan^2 x dx$$

For practice if you want

Theorem

The antiderivatives of rational function are always elementary functions.

Here, rational functions are in the form that $\frac{p_m(x)}{q_n(x)}$, where $p_m(x)$ and $q_n(x)$ are polynomials in the degree of m and n .



Basic rational functions

$$\textcircled{1} \int \frac{dx}{ax+b} = \frac{1}{a} \ln|ax+b| + C$$

$$\textcircled{2} \int \frac{dx}{(ax+b)^n} = \int \frac{d(ax+b)}{a(ax+b)^n} = \frac{1}{a(n-1)(ax+b)^{n-1}} + C$$

$$\textcircled{3} \int \frac{xdx}{ax^2+bx+c} = \frac{1}{2a} \ln|ax^2+bx+c| - \frac{b}{2a} \int \frac{dx}{ax^2+bx+c}$$

$$\textcircled{4} \int \frac{dx}{ax^2+bx+c} \quad \Delta=b^2-4ac<0 \quad \frac{2}{\sqrt{-\Delta}} \arctan \frac{2ax+b}{\sqrt{4ac-b^2}} + C$$

$$\textcircled{5} \int \frac{dx}{ax^2+bx+c} \quad \Delta=b^2-4ac>0 \quad \frac{1}{\sqrt{\Delta}} \ln \left| \frac{2ax+b-\sqrt{\Delta}}{2ax+b+\sqrt{\Delta}} \right| + C$$

Remark. What if $\Delta = 0$?



Partial Fraction (Very Important)

Partial fraction is a useful tool when you solving a problem relating formal power series. In the end, you will always get a rational function. And then you will need partial fraction the separate it and get the coefficients.

For a fraction $\frac{P(X)}{Q(X)}$ with $\deg P < \deg Q$ (partial fraction is valid under this condition),

Factor in denominator	Term in partial fraction decomposition
$ax + b$	$\frac{A}{ax+b}$
$(ax + b)^k$	$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_k}{(ax+b)^k}, k \in \mathbb{N}^*$
$ax^2 + bx + c$	$\frac{Ax+B}{ax^2+bx+c}$
$(ax^2 + bx + c)^k$	$\frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \cdots + \frac{A_kx+B_k}{(ax^2+bx+c)^k}, k \in \mathbb{N}^*$

First, you need to factorize $Q(X)$ into the product of irreducible terms. Since we work on $\mathbb{R}$, the degree of an irreducible term would be at most 2. Then, determine the partial fraction term according to the table. Finally, get those constants via evaluating x , solving system of equations or other techniques.



EXAMPLE 6 Evaluate $\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx$.

EXAMPLE 8 Evaluate $\int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} dx$.

$$b+c = -2$$

$$x=1$$

$$-1 - \frac{1}{4} + \frac{c-b}{2} = -\frac{1}{4}$$

$$b=2$$

$$b+c=-1$$

$$\frac{1}{x} = 1 + \frac{1}{x} + \frac{b+c}{2}$$

$$\textcircled{-1}$$

$$\frac{1}{x} + \frac{x}{(x^2+1)^2} + \frac{bx+c}{x^2+1}$$

$$\textcircled{x=i}$$

$$\frac{1-i-2+i}{i} \textcircled{1} = D i + E$$

$$E=0$$

$$D=1$$

$$\frac{A}{x} + \frac{bx+c}{x^2+1} + \frac{dx+e}{(x^2+1)^2}$$

$$\textcircled{x=1}$$

$$\textcircled{A=1}$$

$$2+2+1$$

$$\frac{5}{-4}$$

$$-1 \times 4$$

$$\frac{1}{x} + \frac{-x-1}{x^2+1} + \frac{x}{(x^2+1)^2}$$

SOLUTION Since the degree of the numerator is *not less than* the degree of the denominator, we first divide and obtain

$$\frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} = 1 + \frac{x - 1}{4x^2 - 4x + 3}$$

Notice that the quadratic $4x^2 - 4x + 3$ is irreducible because its discriminant is $b^2 - 4ac = -32 < 0$. This means it can't be factored, so we don't need to use the partial fraction technique.

To integrate the given function we complete the square in the denominator:

$$4x^2 - 4x + 3 = (2x - 1)^2 + 2$$

This suggests that we make the substitution $u = 2x - 1$. Then $du = 2 \, dx$ and $x = \frac{1}{2}(u + 1)$, so

$$\begin{aligned} \int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} \, dx &= \int \left(1 + \frac{x - 1}{4x^2 - 4x + 3} \right) \, dx \\ &= x + \frac{1}{2} \int \frac{\frac{1}{2}(u + 1) - 1}{u^2 + 2} \, du = x + \frac{1}{4} \int \frac{u - 1}{u^2 + 2} \, du \\ &= x + \frac{1}{4} \int \frac{u}{u^2 + 2} \, du - \frac{1}{4} \int \frac{1}{u^2 + 2} \, du \\ &= x + \frac{1}{8} \ln(u^2 + 2) - \frac{1}{4} \cdot \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{u}{\sqrt{2}} \right) + C \\ &= x + \frac{1}{8} \ln(4x^2 - 4x + 3) - \frac{1}{4\sqrt{2}} \tan^{-1} \left(\frac{2x - 1}{\sqrt{2}} \right) + C \end{aligned}$$



SOLUTION The form of the partial fraction decomposition is

$$\frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{(x^2 + 1)^2}$$

Multiplying by $x(x^2 + 1)^2$, we have

$$\begin{aligned} -x^3 + 2x^2 - x + 1 &= A(x^2 + 1)^2 + (Bx + C)x(x^2 + 1) + (Dx + E)x \\ &= A(x^4 + 2x^2 + 1) + B(x^4 + x^2) + C(x^3 + x) + Dx^2 + Ex \\ &= (A + B)x^4 + Cx^3 + (2A + B + D)x^2 + (C + E)x + A \end{aligned}$$

If we equate coefficients, we get the system

$$A + B = 0 \quad C = -1 \quad 2A + B + D = 2 \quad C + E = -1 \quad A = 1$$

which has the solution $A = 1$, $B = -1$, $C = -1$, $D = 1$, and $E = 0$. Thus

$$\begin{aligned} \int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} dx &= \int \left(\frac{1}{x} - \frac{x + 1}{x^2 + 1} + \frac{x}{(x^2 + 1)^2} \right) dx \\ &= \int \frac{dx}{x} - \int \frac{x}{x^2 + 1} dx - \int \frac{dx}{x^2 + 1} + \int \frac{x dx}{(x^2 + 1)^2} \\ &= \ln |x| - \frac{1}{2} \ln(x^2 + 1) - \tan^{-1}x - \frac{1}{2(x^2 + 1)} + K \end{aligned}$$



Exercise 7-Reciprocal Substitution

$$① \int \frac{dx}{x^8(1+x^2)}$$

Exercise 8-Rationalization

$$① \int \frac{dx}{\sqrt{1-x} + \sqrt[3]{1-x}}$$

Ex 7.

$$\int \frac{dx}{x^3(4x^2+3)}$$

let $u = \frac{1}{x}$ $du = -\frac{1}{x^2} dx \Rightarrow dx = -x^2 du$

$$\begin{aligned} \int \frac{dx}{x^3(4x^2+3)} &= \int \frac{-u^3 du}{\frac{u^3}{u^2+1}} = - \int \frac{u^3}{u^2+1} du = - \int \frac{u^2+1}{u^2+1} du = - \int \frac{u^2+1}{u^2+1} du + \int \frac{1}{u^2+1} du \\ &= - \int (u^2 + 1) du + \arctan u \\ &= -\frac{1}{3}u^3 - u + \arctan u \\ &\rightarrow \text{transform } u \text{ back to } x \end{aligned}$$

Triangle Substitution



Ex. 8.

$$\int \frac{dx}{\sqrt{x+3}\sqrt{1-x}} \quad \text{let } u = \sqrt{1-x} \quad du = \frac{1}{2}(1-x)^{-\frac{1}{2}} \cdot -1 = -\frac{1}{2}(1-x)^{-\frac{1}{2}} dx = -\frac{1}{2}u^{-1} dx \Rightarrow -2u du = dx$$

$$= \int \frac{-2u^5}{u^3+u^2} du = -2 \int u^2 \cdot \frac{u^4+u^2+1}{u^4+u^2} du$$

$$= -2 \int \frac{u^5}{u^3+u^2} du = -2 \int u^2 - u + \frac{u^3}{u^3+u^2} du$$

$$= -2 \int \frac{u^5+u^4-u^4}{u^3+u^2} du \quad \downarrow \text{same as above}$$

$$= -2 \int u^2 - \frac{u^4}{u^3+u^2} du$$

When encountering complex combination of trigonometric functions, we can substitute $t = \tan \frac{x}{2}$.

Then there is

$$\sin x = \frac{2t}{t^2 + 1}, \cos x = \frac{1 - t^2}{t^2 + 1}, \tan x = \frac{2t}{1 - t^2}, \frac{dx}{dt} = \frac{2}{t^2 + 1}.$$

It is thus converted into rational functions.

Exercise 9-"Universal Substitution"

$$\textcircled{1} \int \frac{dx}{1 + \cos x + \sin^2\left(\frac{x}{2}\right)}$$

Improper Integral

Definition

Type1: Unboundedness

- ① If $f(x)$ is continuous on $[a, +\infty]$, for any $t > a$, the improper integral of $f(x)$ on $[a, +\infty]$ is

$$\int_a^{+\infty} f(x)dx = \lim_{t \rightarrow +\infty} \int_a^t f(x)dx$$

- ② If $f(x)$ is continuous on $[-\infty, a]$, for any $t < a$, the improper integral of $f(x)$ on $[-\infty, a]$ is

$$\int_a^{+\infty} f(x)dx = \lim_{t \rightarrow -\infty} \int_t^a f(x)dx$$



Improper Integral

Definition

Type1: Unboundedness

- ③ If both $\int_a^{\infty} f(x)dx$ and $\int_{-\infty}^a f(x)dx$ are convergent

$$\int_{-\infty}^{+\infty} f(x)dx = \int_a^{+\infty} f(x)dx + \int_{-\infty}^a f(x)dx$$



Improper Integral

Definition

Type2: Discontinuous

- ① If f is continuous on $[a,b)$ and is discontinuous at b , then the improper integral on is

$$\int_a^b f(x)dx = \lim_{t \rightarrow b^-} \int_a^t f(x)dx$$

- ② If f is continuous on $(a,b]$ and is discontinuous at a , then the improper integral on is

$$\int_a^b f(x)dx = \lim_{t \rightarrow a^+} \int_t^b f(x)dx$$



~

Comparison Theorem Suppose that f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$.

(a) If $\int_a^\infty f(x) dx$ is convergent, then $\int_a^\infty g(x) dx$ is convergent.

(b) If $\int_a^\infty g(x) dx$ is divergent, then $\int_a^\infty f(x) dx$ is divergent.

Problem type1

V **EXAMPLE 6** Determine whether $\int_0^{\pi/2} \sec x \, dx$ converges or diverges.

V **EXAMPLE 9** Show that $\int_0^{\infty} e^{-x^2} dx$ is convergent.

$$\begin{aligned}\int_0^{\pi/2} \sec x \, dx &= \lim_{t \rightarrow (\pi/2)^-} \int_0^t \sec x \, dx = \lim_{t \rightarrow (\pi/2)^-} \ln |\sec x + \tan x| \Big|_0^t \\ &= \lim_{t \rightarrow (\pi/2)^-} [\ln(\sec t + \tan t) - \ln 1] = \infty\end{aligned}$$

because $\sec t \rightarrow \infty$ and $\tan t \rightarrow \infty$ as $t \rightarrow (\pi/2)^-$. Thus the given improper integral is divergent.

SOLUTION We can't evaluate the integral directly because the antiderivative of e^{-x^2} is not an elementary function (as explained in Section 7.5). We write

$$\int_0^{\infty} e^{-x^2} dx = \int_0^1 e^{-x^2} dx + \int_1^{\infty} e^{-x^2} dx$$

and observe that the first integral on the right-hand side is just an ordinary definite integral. In the second integral we use the fact that for $x \geq 1$ we have $x^2 \geq x$, so $-x^2 \leq -x$ and therefore $e^{-x^2} \leq e^{-x}$. (See Figure 13.) The integral of e^{-x} is easy to evaluate:

$$\int_1^{\infty} e^{-x} dx = \lim_{t \rightarrow \infty} \int_1^t e^{-x} dx = \lim_{t \rightarrow \infty} (e^{-1} - e^{-t}) = e^{-1}$$

Thus, taking $f(x) = e^{-x}$ and $g(x) = e^{-x^2}$ in the Comparison Theorem, we see that $\int_1^{\infty} e^{-x^2} dx$ is convergent. It follows that $\int_0^{\infty} e^{-x^2} dx$ is convergent. ■

In Example 9 we showed that $\int_0^{\infty} e^{-x^2} dx$ is convergent without computing its value. In Exercise 70 we indicate how to show that its value is approximately 0.8862. In probability theory it is important to know the exact value of this improper integral, as we will see in Section 8.5; using the methods of multivariable calculus it can be shown that the exact value is $\sqrt{\pi}/2$. Table 1 illustrates the definition of an improper integral by showing how the (computer-generated) values of $\int_0^t e^{-x^2} dx$ approach $\sqrt{\pi}/2$ as t becomes large. In fact, these values converge quite quickly because $e^{-x^2} \rightarrow 0$ very rapidly as $x \rightarrow \infty$.



Problem type2

2

$\int_1^{\infty} \frac{1}{x^p} dx$ is convergent if $p > 1$ and divergent if $p \leq 1$.

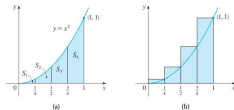
2 Definition of a Definite Integral If f is a function defined for $a \leq x \leq b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$. We let $x_0 (= a)$, $x_1, x_2, \dots, x_n (= b)$ be the endpoints of these subintervals and we let $x_1^*, x_2^*, \dots, x_n^*$ be any **sample points** in these subintervals, so x_i^* lies in the i th subinterval $[x_{i-1}, x_i]$. Then the **definite integral of f from a to b** is

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

provided that this limit exists and gives the same value for all possible choices of sample points. If it does exist, we say that f is **integrable** on $[a, b]$.

EXAMPLE 1 Use rectangles to estimate the area under the parabola $y = x^2$ from 0 to 1 (the parabolic region S illustrated in Figure 3).

SOLUTION We first notice that the area of S must be somewhere between 0 and 1 because S is contained in a square with side length 1, but we can certainly do better than that. Suppose we divide S into four strips S_1, S_2, S_3 , and S_4 by drawing the vertical lines $x = \frac{1}{4}$, $x = \frac{2}{4}$, and $x = \frac{3}{4}$ as in Figure 4(a).



Example

$$① \lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{i}{n^2}$$

$$② \lim_{n \rightarrow \infty} \sqrt{n} \left(\frac{1}{n} - \sum_{i=0}^n \frac{1}{n + \sqrt{i}} \right)$$



Method

① Simply Transform into Integration. $\lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{i}{n^2} = \int_0^n \frac{i}{n^2} di = \frac{1}{2}$

② Use Squeeze Theorem. $ans = \lim_{n \rightarrow \infty} \sqrt{n} \sum_{i=0}^n \frac{\sqrt{i}}{n(n + \sqrt{i})}$

$$ans \leq \lim_{n \rightarrow \infty} \sqrt{n} \sum_{i=0}^n \frac{\sqrt{i}}{n^2} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^n \sqrt{\frac{k}{n}} = \int_0^1 \sqrt{x} dx = \frac{2}{3}.$$

$$ans \geq \lim_{n \rightarrow \infty} \sqrt{n} \sum_{i=0}^n \frac{\sqrt{i}}{n(n + \sqrt{n})} = \lim_{n \rightarrow \infty} \frac{n}{n + \sqrt{n}} \frac{1}{n} \sum_{i=0}^n \sqrt{\frac{k}{n}} = \frac{2}{3}.$$

Exercise: Refer to Worksheet 1 1.3,1.4



Reference

- James Stewart Calculus Early Transcendentals 7th Edition
- VV156 24FA Lecture Slides, Runze Cai

THANK YOU AND GOOD LUCK!